



Course Code: CL-2005	Course: Database Systems Lab
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Group by Statement:

The GROUP BY statement groups rows that have the same values in summary rows, like “Find the number of customers in each country”.

The GROUP BY statement is often used with aggregate functions (COUNT, MAX, MIN, SUM, AVG) to group the result-set by one or more columns.

Group by Syntax

SELECT column name(s) **FROM** table name **GROUP BY** column name(s)

```
SELECT
    AVG(salary) as "average_salary"
FROM
    employees
GROUP BY Department id
```

Sample Output:

[illegible]

Group by (Having)

Having Clause is used with GROUP BY clause to restrict the groups of returned rows where condition is TRUE.

Syntax:

```
SELECT expression1, expression2, ... expression_n,  
    aggregate_function (aggregate_expression)  
FROM table_name  
WHERE conditions  
GROUP BY expression1, expression2, ... expression_n  
HAVING having_condition;
```

HAVING Example: (with GROUP BY SUM function)

```

SELECT item, SUM(sale) AS "Total sales"
FROM salesdepartment
GROUP BY item
HAVING SUM(sale) < 1000;

```

HAVING Example: (with GROUP BY MIN function)

```

SELECT Department_ID,
MIN(salary) AS "Lowest salary"
FROM employees
GROUP BY Department_ID
HAVING MIN(salary) < 15000;

```

Sample Output:

	DEPARTMENT_ID	Lowest salary
1	100	6900
2	30	2500
3	(null)	7000
4	20	6000
5	70	10000
6	110	8300
7	50	2100
8	80	6100
9	40	6500
10	60	4200
11	10	4400

HAVING Example: (with GROUP BY MAX function)

```

SELECT Department_ID,
MAX(salary) AS "Highest salary"
FROM employees
GROUP BY Department_ID
HAVING MAX(salary) > 3000;

```

Sample Output:

	DEPARTMENT_ID	Highest salary
1	100	12008
2	30	11000
3	(null)	7000
4	90	24000
5	20	13000
6	70	10000
7	110	12008
8	50	8200
9	80	14000
10	40	6500
11	60	9000
12	10	4400

Sub Queries:

A Subquery is a query within another SQL query and embedded within the WHERE clause.

Important Rule:

- A subquery can be placed in a number of SQL clauses like WHERE clause, FROM clause, HAVING clause.
- You can use Subquery with SELECT, UPDATE, INSERT, DELETE statements along with the operators like =, <, >, >=, <=, IN, BETWEEN, etc.
- A subquery is a query within another query. The outer query is known as the main query, and the inner query is known as a subquery.
- Subqueries are on the right side of the comparison operator.
- A subquery is enclosed in parentheses.
- In the Subquery, ORDER BY command cannot be used. But GROUP BY command can be used to perform the same function as ORDER BY command.

NOTE:

Subqueries are useful when a query is based on unknown values.

Sub Queries with SELECT Statement:Syntax:

```
SELECT column_name
FROM table_name
WHERE column_name expression operator
( SELECT column_name FROM table_name WHERE ... );
```

Types of Subqueries:

1. **Single Row Sub Query:** Sub query which returns single row output. They mark the usage of single row comparison operators, when used in WHERE conditions.
2. **Multiple row sub query:** Sub query returning multiple row output. They make use of multiple row comparison operators like IN, ANY, ALL. There can be sub queries returning multiple columns also.
3. **Correlated Sub Query:** Correlated subqueries depend on data provided by the outer query. This type of subquery also includes subqueries that use the EXISTS operator to test the existence of data rows satisfying specified criteria.

Single Row Sub Queries:

- Return only one row
- Use single-row comparison operators

Operator	Meaning
=	Equal to
>	Greater than
>=	Greater than or equal to
<	Less than
<=	Less than or equal to
<> , !=	Not equal to

```
SELECT First_Name, Job_ID FROM Employees WHERE job = ( SELECT job_ID FROM
Employees WHERE empno=7369 )
```

Sample Output:

	FIRST_NAME	JOB_ID
1	Alexander	PU_CLERK
2	Shelli	PU_CLERK
3	Sigal	PU_CLERK
4	Guy	PU_CLERK
5	Karen	PU_CLERK

Single Row Functions:**Finds the employees who have the highest salary:**

```

SELECT
    employee_id, first_name, last_name, salary
FROM
    employees
WHERE
    salary = (SELECT MAX(salary) FROM employees)

```

Sample Output:

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY
1	100	Steven	King	24000

Finds all employees who salaries are greater than the average salary of all employees:

```

SELECT
    employee_id, first_name, last_name, salary
FROM
    employees
WHERE
    salary > (SELECT AVG(salary) FROM employees)

```

Sample Output:

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY
1	100	Steven	King	24000
2	101	Neena	Kochhar	17000
3	102	Lex	De Haan	17000
4	103	Alexander	Hunold	9000
5	108	Nancy	Greenberg	12008
6	109	Daniel	Faviet	9000
7	110	John	Chen	8200
8	111	Ismael	Sciarra	7700
9	112	Jose Manuel	Urman	7800
10	113	Luis	Popp	6900

Multiple row sub query:

- Return more than one row
- Use multiple-row comparison operators
 - [**>** ALL] More than the highest value returned by the subquery
 - [**<** ALL] Less than the lowest value returned by the subquery
 - [**<** ANY] Less than the highest value returned by the subquery
 - [**>** ANY] More than the lowest value returned by the subquery
 - [**=** ANY] Equal to any value returned by the subquery (same as IN)

IN:

```

SELECT first_name, department_id
FROM employees
WHERE department_id IN (SELECT Department_id
FROM departments
WHERE LOCATION_ID = 100)

```

Sample Output:

	FIRST_NAME	DEPARTMENT_ID
1	Shelli	30
2	John	100
3	Karen	30
4	Lex	90
5	Daniel	100
6	William	110
7	Nancy	100
8	Shelley	110
9	Guy	30
10	Alexander	30

ANY:

```

SELECT employee_ID, First_Name, job_ID
FROM EMPLOYEES
WHERE SALARY < ANY
( SELECT salary FROM EMPLOYEES WHERE JOB_ID = 'PU_CLERK' );

```

Sample Output:

	EMPLOYEE_ID	FIRST_NAME	JOB_ID
1	132	TJ	ST_CLERK
2	128	Steven	ST_CLERK
3	136	Hazel	ST_CLERK
4	127	James	ST_CLERK
5	135	Ki	ST_CLERK
6	119	Karen	PU_CLERK
7	131	James	ST_CLERK
8	140	Joshua	ST_CLERK
9	144	Peter	ST_CLERK
10	182	Martha	SH_CLERK

ALL:

```

SELECT employee_ID, First_Name, job_ID
FROM EMPLOYEES WHERE SALARY > ALL
( SELECT salary FROM HR.EMPLOYEES WHERE JOB_ID = 'PU_CLERK' ) AND
job_ID <> 'PU_CLERK';

```

Sample Output:

	EMPLOYEE_ID	FIRST_NAME	JOB_ID
1	190	Winston	SH_CLERK
2	125	Julia	ST_CLERK
3	194	Samuel	SH_CLERK
4	138	Stephen	ST_CLERK
5	133	Jason	ST_CLERK
6	129	Laura	ST_CLERK
7	186	Julia	SH_CLERK
8	141	Trenna	ST_CLERK
9	189	Jennifer	SH_CLERK
10	137	Renske	ST_CLERK

Group By and HAVING IN SUB QUERIES:

```

SELECT department_name, avg(salary)
FROM EMP_DETAILS_VIEW
GROUP BY department_name
HAVING avg(salary) > ( SELECT avg(salary) FROM EMPLOYEES);

```

Sample Output:

[illegible]

SUBQUERIES AND DML:

Subqueries with the INSERT Statement

- SQL subquery can also be used with the Insert statement. In the insert statement, data returned from the subquery is used to insert into another table.
- In the subquery, the selected data can be modified with any of the character, date functions.

Syntax:

```
INSERT INTO table_name (column1, column2, column3. )
SELECT *
FROM table_name
WHERE VALUE OPERATOR
```

You may login from a new user for DML sub Queries.

Example: Let's assume we have an EMPLOYEE_BKP table available which is backup of EMPLOYEE table having all the attributes of Employees table

```
INSERT INTO EMPLOYEE_BKP  
SELECT * FROM EMPLOYEES  
WHERE job_ID IN (SELECT job_id  
FROM jobs WHERE job_title='Accountant');
```

Subqueries with the UPDATE Statement

The subquery of SQL can be used in conjunction with the Update statement. When a subquery is used with the Update statement, then either single or multiple columns in a table can be updated.

Syntax

```
UPDATE table
SET column_name = new_value
WHERE VALUE OPERATOR
  (SELECT COLUMN_NAME
   FROM TABLE_NAME
   WHERE condition);
```

Example:

The given example updates the **SALARY** by 10 times in the **EMPLOYEE** table for all employee whose minimum salary is 3000.

```
Update employees
set salary= salary+(0.1*salary)
WHERE job_ID IN (SELECT job_ID
FROM jobs WHERE min salary=3000);
```

Subqueries with the DELETE Statement

The subquery of SQL can be used in conjunction with the Delete statement just like any other statement mentioned above.

Syntax

```
DELETE FROM TABLE_NAME  
WHERE VALUE OPERATOR  
  (SELECT COLUMN_NAME  
   FROM TABLE_NAME  
   WHERE condition);
```

Example:

Let's assume we have an EMPLOYEE_BKP table available which is a backup of EMPLOYEE table. The given example deletes the records from the EMPLOYEE_BKP table for all EMPLOYEE whose end date is '31-DEC-06'.

```
Delete from employee_BKP  
WHERE job_ID IN (SELECT job_ID  
 FROM job_History WHERE end_Date='31-Dec-06');
```

```
SELECT  
  e.employee_id,  
  e.first_name,  
  e.last_name,  
  (SELECT job_title FROM jobs WHERE job_id = e.job_id) AS job_title,  
  (SELECT department_name FROM departments WHERE department_id = e.department_id)  
AS department_name,  
  (SELECT city FROM locations WHERE location_id = d.location_id) AS  
department_location,  
  (SELECT region_name FROM regions WHERE region_id = r.region_id) AS region_name  
FROM  
  employees e,  
  departments d,  
  locations l,  
  regions r  
WHERE  
  e.department_id = d.department_id  
  AND d.location_id = l.location_id;
```

(ROWNUM) LIMIT Function:

In SQL databases, limit function is used to restrict the number of rows returned by a query. Here's a simple explanation of how LIMIT function works:

Example:

Display only the top 5 highest salaries from an employee's table

```
SELECT salary FROM (  
  SELECT salary FROM employees  
  ORDER BY salary DESC  
)  
WHERE ROWNUM <= 5;
```

Lab Tasks:

1. Write a query to display the employee name (first name and last name) whose salary is greater than the salary of any employee working in department 80.
2. Write a query to display the first name, last name, and hire date of those employees who were hired before the employee with ID = 150.
3. Write a query to display the job title of the employee who earns the maximum salary in the HR schema.
4. Write a query to display all employees who work in the same department as the employee named “Neena”.
5. Write a query to display the department name where the highest-paid employee in the company is working.
6. Write a query to display the employee name and salary for all employees whose salary is greater than the **average salary** of their department.
7. Write a query to display the first name, last name, and department ID of employees who work in the same department as the employee with the lowest salary.
8. Write a query to display the names of employees who earn less than the salary of “Lex De Haan”.
9. Write a query to display the department name and manager ID of the department managed by the manager who earns the lowest salary among all managers.
10. Write a query to display the employee ID, name, and salary of employees who earn more than the **average salary of all employees hired after 01-JAN-2005**.
11. Write a query to display the department ID, department name, and location ID of those departments that are located in the same city as department 20.
12. Write a query to display the name and hire date of all employees hired after the most recently hired employee in department 90.
13. Write a query to display the employee IDs of employees who are not managers (use subquery on the managers’ list).
14. Write a query to display the name, salary, and department ID of employees who earn the **second highest salary** in their department.
15. Write a query to display the department names which have more employees than the department where employee ‘David’ works.
16. Write a query to display the employee name and job title for those who have the same job as the employee with ID = 176.
17. Write a query to display the city where the employee who earns the minimum salary works.
18. Write a query to display the names of employees who earn more than the average salary of employees in department 60.
19. Write a query to display the employee IDs and names of employees who work in departments that do not have any manager assigned.
20. Write a query to display the department name where the maximum number of employees are working.